

Supply voltage with disconnected plug Airbag sensor and connected Plug control module Airbag

- Ignition on
- Measurement/s at the plug contacts **Airbag sensor** must be performed (Plug disconnected)
- Nominal value : 2.0 V $\Rightarrow$ 2.5 V (measured value fluctuates by +/- 0.15 V)

Supply voltage with connected plug Airbag sensor and connected Plug control module Airbag

- Ignition on
- Measurement/s at the plug contacts **Airbag sensor**
- Nominal value : 8.0 V $\Rightarrow$ 8.2 V

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test without connected components

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V $\Rightarrow$ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

Fault effects

- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

Diagnosis information - Airbag

Airbag pressure sensor Passenger's door

⚠ WARNING

Uncontrolled triggering of the airbag

- **Contusions, crushing injuries or death**

- ▶ **Never hang the PIWIS Tester** on or over the steering wheel.
- ▶ Before working on the ignition circuits of the airbag system, always disconnect the battery from the vehicle electrical system and ensure that the vehicle electrical system is de-energized.
- ▶ Wait for at least one minute before starting work on the airbag system.
- ▶ Perform fault finding on the ignition circuits of the airbag system and its lines only using approved special tools.

⚠ WARNING

Driving with attached/connected special tools or commercially available tools

- **Contusions, crushing injuries or death**
- **Traffic hazard**

- ▶ Never drive with special tools and/ or commercially available tools attached instead of the airbag units.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- **Supply voltage** > 9.1 V
- Ignition on > 5 Second(s), then:
- Ignition off > 2 Second(s), then:
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- Airbag triggering unit Internal function test

Possible fault causes

- ♦ **Coding** - Ignition circuit is connected and OK, but is not coded.
- ♦ Electrical fault in circuit - Electrical connection between two ignition circuits (mutual short circuit). Short circuit in wiring harness of two ignition circuits
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Front pedestrian protection sensor Driver's side

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- ♦ **Airbag sensor** Faulty

Supply voltage with disconnected plug Airbag sensor and connected Plug control module Airbag

- Ignition on
- Measurement/s at the plug contacts **Airbag sensor** must be performed (Plug disconnected)
- Nominal value : 2.0 V $\Rightarrow$ 2.5 V (measured value fluctuates by +/- 0.15 V)

Supply voltage with connected plug Airbag sensor and connected Plug control module Airbag

- Ignition on
- Measurement/s at the plug contacts **Airbag sensor**
- Nominal value : 8.0 V $\Rightarrow$ 8.2 V

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test without connected components

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V $\Rightarrow$ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

Fault effects

- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

Diagnosis information - Airbag

Front pedestrian protection sensor Passenger's side

⚠ WARNING

Uncontrolled triggering of the airbag

- **Contusions, crushing injuries or death**

- ▶ **Never hang the PIWIS Tester** on or over the steering wheel.
- ▶ Before working on the ignition circuits of the airbag system, always disconnect the battery from the vehicle electrical system and ensure that the vehicle electrical system is de-energized.
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⚠ WARNING

Driving with attached/connected special tools or commercially available tools

- **Contusions, crushing injuries or death**
- **Traffic hazard**

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Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- **Supply voltage** > 9.1 V
- Ignition on > 5 Second(s), then:
- Ignition off > 2 Second(s), then:
- Ignition on > 5 Second(s)
- Airbag triggering unit Internal function test

Possible fault causes

- ♦ **Coding** - Ignition circuit is connected and OK, but is not coded.
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Diagnosis information - Airbag

Electrical fault Control module Airbag

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- Ignition on
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Possible fault causes

- ♦ Short circuit to ground (Electrical fault in circuit)
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- ♦ Short circuit between lines

Fault effects

- Crash signal cannot be output (fault 00490x)
- More ignition circuits than required may be triggered in the event of a crash (faults 004E03, 004D03)

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Electrical fault Control module Airbag

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Diagnosis information - Airbag

Coding

Diagnostic conditions

- Ignition on
- Airbag triggering unit Internal function test

Possible fault causes

- ♦ Airbag triggering unit incorrectly coded
- ♦ Vehicle equipment incorrect or incomplete